PAPER_429 – Three New UQFF Number Systems: Vacuum Density Series, Dipole Vortex Primes, Buoyancy Harmonics

Unified Quantum Field Framework (UQFF)

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Source: grok_share_c020496d9e.txt — Clarification sections and Vacuum Density Series formulae (lines 800–880 and lines ~224–237, Session 114 deep-physics extraction) **Session:** 114 **CP4 Class:** `ThreeNewNumberSystemsVacuumDipoleBuoyancyCalculator` (#83)

1. Overview

PAPER_429 identifies and formalises **three new number systems** that emerge naturally from the UQFF mathematical framework — analogous to Ramanujan series, Bernoulli numbers, and harmonic series in classical mathematics, but rooted in 26-dimensional vacuum physics. Each system encodes a different aspect of the UQFF buoyancy-magnetism structure.

2. Number System I: Vacuum Density Series

2.1 Definition

$$\boxed{V_n = \sum_{k=1}^{\infty} \frac{[\text{SSq}]^k}{k^{26}}}$$

This is a polylogarithm-type series: $V_n = \text{Li}_{26}([\text{SSq}])$ evaluated at the UQFF superconducting medium index.

2.2 Convergence

The series converges absolutely for $|[\text{SSq}]| < 1$. Since $[\text{SSq}] = 0.57 < 1$, the series is well-defined. The exponent 26 is not arbitrary — it equals the number of UQFF dimensional layers from PAPER_427.

2.3 Physical Interpretation

Each term $[\text{SSq}]^k / k^{26}$ represents the vacuum density contribution from the k-th excitation level of the SCm field projected through all 26 dimensional channels simultaneously. The series sum gives the **total vacuum energy partition** accessible to the buoyancy field.

2.4 Numerical Values

With $[\text{SSq}] = 0.57$:

k	$[\text{SSq}]^k / k^{26}$	Cumulative sum
1	5.700×10^{-1}	0.570
2	7.688×10^{-9}	0.570
3	7.278×10^{-14}	~0.570
∞	—	≈ 0.5700

The series is dominated by $k=1$; higher terms contribute less than 10^{-8} due to the k^{26} denominator.

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3. Number System II: Dipole Vortex Primes

3.1 Definition

The **Dipole Vortex Prime sequence** $\mathcal{P}_{\text{DV}}$ consists of the prime numbers $p > 26$:

$$\mathcal{P}_{\text{DV}} = \{29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, \ldots\}$$

The 27th prime is $p_{27} = 103$ and the 30th prime is $p_{30} = 113$.

3.2 Physical Interpretation

Each prime $p \in \mathcal{P}_{\text{DV}}$ encodes a **U_g3 magnetic string vortex state**: the prime gap structure determines the phase separation between adjacent vortex lines in the SCm vacuum.

The vortex energy for the n-th Dipole Vortex Prime is:

$$E_{\text{vortex}}(p_n) = \frac{\hbar \omega_{\text{str}}}{p_n} \cdot \phi^{p_n \mod 6}$$

where $\phi = (1+\sqrt{5})/2$ is the golden ratio.

3.3 Special Prime: $p_{\text{special}} = 113$

$$p_{\text{special}} = 113 \quad \text{(30th prime, first prime after the 26-layer Hydrogen proto-shell)}$$

This prime marks the **hydrogen proto-shell anchor**: the period-4 shell ($Z=1$ to $Z=18$, with 18 electrons = $2 + 8 + 8$) terminates at a structure whose dimension is captured by prime 113. In the UQFF string topology, $p=113$ is the vortex index at which the fundamental U_g3 string completes a full 26D topology cycle.

3.4 Connection to Ug3

The magnetic string rotation term U_{g3} depends on the Dipole Vortex Prime spectrum:

$$U_{g3}(t) = \sum_{n: p_n > 26} \frac{A_{\text{str}}}{p_n} \cdot \cos\!\left(\omega_{\text{str}} p_n \cdot t + \varphi_{p_n}\right)$$

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4. Number System III: Buoyancy Harmonics

4.1 Definition

The m-th **Buoyancy Harmonic** H_m is an UQFF analogue of the harmonic series:

$$H_m = \sum_{k=1}^{m} \frac{f_{\text{Ub}}}{k}, \qquad f_{\text{Ub}} = k_{\text{Ub}} \cdot \Delta k_\eta \cdot \frac{\rho_{\text{vac,UA}}}{\rho_{\text{vac,SCm}}} \cdot \frac{\Delta \rho}{\rho_{\text{UA}}}$$

4.2 U_g2 Buoyancy Harmonic Sum

The total Ug2 buoyancy field is the sum over all harmonics:

$$\boxed{U_{g2}(t) = \sum_{m=1}^{\infty} H_m \cdot \left(1 - e^{-[\text{SSq}]\, m}\right) \cdot \cos\!\left(\omega_{U_{g2}} \cdot t_n\right)}$$

4.3 Convergence

Unlike the classical harmonic series $\sum 1/k$ (which diverges), the Buoyancy Harmonic sum for U_{g2} converges because the $(1 - e^{-[\text{SSq}]m})$ factor grows toward 1 while the term $H_m \cdot e^{-[\text{SSq}]m} = \left(\sum_{k=1}^m f/k\right) e^{-[\text{SSq}]m} \to 0$ for large m.

4.4 Physical Interpretation

- H_m grows logarithmically with m — corresponding to the logarithmic buildup of buoyancy modes

- The factor $(1 - e^{-[\text{SSq}]m})$ ensures **vacuum saturation**: once the SCm medium has absorbed all m buoyancy modes, additional modes contribute negligibly
- $\cos(\omega_{U_{g2}} t_n)$ is the time-oscillatory projection onto the negative-time parameter

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5. Dynamic [SSq] Formula

PAPER_429 also identifies the **dynamic [SSq] formula** — replacing the static calibration constant $[\text{SSq}] = 0.57$ with a time- and mode-dependent expression:

$$\boxed{[\text{SSq}](n, t) = \log\!\left(\frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}'}\right) \cdot n \cdot e^{-(\pi - t)}}$$

where ρ_{UA}' is the reduced UA density after SCm phase transition.

At $n=1$, $t=0$: $[\text{SSq}](1,0) = \log(0.1) \cdot 1 \cdot e^{-\pi} \approx (-2.303)(0.0432) \approx -0.0995$ — the dynamic value is approximately 10× smaller than the static calibration, consistent with early-universe conditions.

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6. Relationships Between the Three Systems

Property	Vacuum Series	Dipole Primes	Buoyancy Harmonics
Index domain	$k = 1, 2, 3, \ldots$	primes $p > 26$	$m = 1, 2, 3, \ldots$
Convergence	Polylogarithm Li_{26}	Prime series (conditional)	Modified harmonic
Layer exponent	k^{26} — all 26 dims	$p > 26$ — post-26D primes	[SSq]·m — SCm saturation
Physical field	U_g1 vacuum energy	U_g3 string rotation	U_g2 charge reactivity

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7. Relation to Other Papers

PAPER	Relation
PAPER_427	26D layer count = exponent in Vacuum Series denominator (k^{26})
PAPER_428	$p_{\text{special}}=113$ anchors the hydrogen proto-shell

PAPER	Relation
PAPER_426	Dynamic [SSq] formula replaces static 0.57 in UTe2 δ_n calculation

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8. CP4 Implementation

Class: `ThreeNewNumberSystemsVacuumDipoleBuoyancyCalculator` **Methods:**

- `compute_vacuum_density_series(SSq, n_max)` → V_n partial sum up to $n_{\max}$
- `get_dipole_vortex_primes(n_max)` → first $n_{\max}$ DV primes ($p > 26$)
- `compute_vortex_energy(p, omega_str, phi)` → $E_{\text{vortex}}(p)$
- `compute_buoyancy_harmonic(m, f_Ub)` → H_m
- `compute_Ug2(t_n, omega_Ug2, SSq, f_Ub, m_max)` → $U_{g2}(t)$ sum
- `compute_SSq_dynamic(n, t, rho_SCm, rho_UA_prime)` → $[\text{SSq}](n,t)$

Extracted from grok_share_c020496d9e.txt lines 800–880 and lines 224–237 (Session 114). Three new Ramanujan-class number systems emerge from the UQFF vacuum structure, encoding the 26D buoyancy field across all known physical domains.